moves the gauge needle to the middle of the normal range on the gauge scale to represent the engine oil pressure. The gauge needle will continue to be positioned at the middle of normal range on the gauge scale until the cluster receives a message from the PCM that indicates the engine oil pressure is low, or until the ignition switch is turned to the Off position, whichever occurs first.

- **Engine Oil Pressure Low Message** - Each time the cluster receives a message from the PCM indicating the engine oil pressure of a 2.4L engine is below about 0.2 kg/cm^2 (3 psi), or of any engine other than a 2.4L is below about 0.4 kg/cm^2 (6 psi), the gauge needle is moved to the graduation at the far left (low) end of the gauge scale, the check gauges indicator is illuminated, and a single chime tone is generated. The gauge needle remains at the left end of the gauge scale and the check gauges indicator remains illuminated until the cluster receives a message from the PCM indicating that the engine oil pressure of a 2.4L engine is above about 0.2 kg/cm^2 (3 psi), or of any engine other than a 2.4L is above about 0.4 kg/cm^2 (6 psi), or until the ignition switch is turned to the Off position, whichever occurs first. The cluster will only turn the check gauges indicator on in response to an engine oil pressure low message if the engine speed message is 300 rpm or greater for more than about five seconds.
- **Communication Error** - If the cluster fails to receive an engine oil pressure message, it will hold the gauge needle at the last indication about twelve seconds or until the ignition switch is turned to the Off position, whichever occurs first. After twelve seconds, the cluster will move the gauge needle to the left end of the gauge scale.
- **Actuator Test** - Each time the cluster is put through the actuator test, the gauge needle will be swept to several calibration points on the gauge scale in a prescribed sequence to confirm the functionality of the gauge and the cluster control circuitry.

The PCM continually monitors the engine oil pressure sensor to determine the engine oil pressure. The PCM then sends the proper engine oil pressure messages to the instrument cluster. For further diagnosis of the oil pressure gauge or the instrument cluster circuitry that controls the gauge, (Refer to 8 - ELECTRICAL/INSTRUMENT CLUSTER - DIAGNOSIS AND TESTING). If the instrument cluster turns on the check gauges indicator due to a low oil pressure gauge reading, it may indicate that the engine or the engine oiling system requires service. For proper diagnosis of the engine oil pressure sensor, the PCM, the PCI data bus, or the electronic message inputs to the instrument cluster that control the oil pressure gauge, a DRBIII® scan tool is required. Refer to the appropriate diagnostic information.

REAR LOCK INDICATOR

DESCRIPTION

Fig. 21 Rear Lock Indicator

A rear lock indicator is standard equipment on all instrument clusters, but is only functional on vehicles equipped with the optional off-road package (Fig. 21). The rear lock indicator is located near the upper edge of the instrument cluster, to the right of center. The rear lock indicator consists of a stencil-like cutout of the words “REAR LOCK” in the opaque layer of the instrument cluster overlay. The dark outer layer of the overlay prevents the indicator from being clearly visible when it is not illuminated. A amber Light Emitting Diode (LED) behind the cutout in the opaque layer of the overlay causes the “REAR LOCK” text to appear in amber through the translucent outer layer of the overlay when the indicator is illuminated from behind by the LED, which is soldered onto the instrument cluster electronic circuit board. The illumination intensity of the rear lock indicator is one-step dimmable. When the exterior lighting is turned On, the indicator is dimmed; and, when the exterior lighting is turned Off, the indicator is illuminated at full intensity. The rear lock indicator is serviced as a unit with the instrument cluster.

OPERATION

The rear lock indicator gives an indication to the vehicle operator of the status of the locker mechanism in the rear axle of vehicles equipped with the optional off-road package. The rear lock indicator can also give an indication when certain faults are detected in the axle locker system. This indicator is controlled by a transistor on the instrument cluster electronic circuit board based upon cluster programming, and hard wired inputs to the cluster from the key-in ignition switch, the axle lock switch, and the rear locker indicator switch. The rear lock indicator Light Emitting Diode (LED) is completely controlled by the instrument cluster logic circuit, and that logic will allow this indicator to operate whenever the instrument cluster receives a battery current input on the fused B(+) circuit. Therefore, the LED can be illuminated regardless of the ignition switch position. The cluster can illuminate the LED solid, or flash it on and off at two different rates, slow or fast. The slow rate is about one flash per second, while the fast rate is about two flashes per second. The LED only illuminates when it is provided a path to ground by the instrument cluster transistor. The instrument cluster will turn on the rear lock indicator for the following reasons: